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Audit Firm Industry Specialization and Client Disclosure Quality

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Abstract. This paper provides evidence that clients select auditors as part of their overall disclosure strategy. We hypothesize that in addition to higher quality audits, industry-specialist audit firms assist clients in enhancing disclosures. We also posit that the choice of an industry-specialist auditor signals a client's intention to provide enhanced disclosures. However, we predict that industry-specialist audit firms are less important in regulated industries where enhanced disclosures add little value. Consistent with our hypotheses, we document a positive association between industry-specialist audit firms and analysts' rankings of disclosure quality in unregulated industries, but no relation in regulated industries.

Keywords: auditor industry specialization, disclosure quality, regulated industries

JEL Classification: M41, M42, L15

The objective of this paper is to provide evidence on the effects of hiring an industry specialist auditor. Specifically, we examine the association between the use of an industry specialist audit firm and the quality of the firm's disclosures. This research contributes to the broader question of how auditor choice impacts, or is associated with, financial reporting quality. Contemporaneous research provides some evidence that industry specialists are associated with higher earnings quality. Balsam et al. (2003) provide evidence that industry specialist are associated with higher earnings response coefficients and Gramling et al. (2000) provide evidence that industry specialists are associated with a stronger association between current earnings and subsequent cash flows. The current paper contributes to this evidence by looking at financial reporting quality more broadly defined.

We examine the association between industry specialization and client disclosure quality as measured by the analysts' disclosure quality evaluations reported in the annual Association for Investment Management and Research (AIMR) Corporate Information Committee Reports. AIMR formed industry committees of buy and sell side analysts to evaluate the disclosure quality of companies within selected industries until 1996. Annual AIMR reports evaluated firm disclosure quality on a number of dimensions including annual published information, quarterly and other published information, and investor relations. While the AIMR rankings likely

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capture quality of earnings as part of their rankings, they provide a much broader assessment of a firm's overall disclosure strategy.

This paper provides evidence that clients select auditors as part of their overall disclosure strategy. The evidence is consistent with both industry-specialists assisting clients in enhancing disclosure quality, and the choice of an industry-specialist signaling a client's decision to provide high quality disclosures. The role of industry-specialists in enhancing disclosure quality is consistent with prior research that documents a strong link between client satisfaction and auditor industry specialization and that clients value auditor assistance that goes beyond basic GAAP (Behn et al., 1997). Industry-specialist audit firms also possess industry specific knowledge and expertise that they can cost effectively use to assist clients in developing industry specific disclosure strategies.

A client's choice of an industry-specialist audit firm can also serve as a signal of enhanced disclosure quality.[1] Not all clients seek higher disclosure quality, despite the potential to lower their costs of capital, as they must balance the benefits of enhanced disclosure with the potential cost of transferring information to competitors. Clients who are concerned about information transfer must consider the potential costs of selecting an audit firm that also audits significant competitors. Audit firms develop an intimate knowledge of their clients' business practices and strategies. This knowledge may spill over to an auditor's other clients in the same industry. Because industry-specialists audit a greater portion of an industry than non-specialists, there is increased risk of information spilling over to competitors. Thus, clients who choose not to provide enhanced disclosures due to the potential cost of transferring knowledge to competitors will also not hire a specialist audit firm.

We expect the association between industry specialization and disclosure quality to be strongest in industries where enhanced disclosure adds the most value, namely in unregulated industries where required disclosures generally do not exceed basic GAAP. In contrast, we predict that industry-specialist audit firms have little impact on disclosure quality in regulated industries with high levels of required disclosure and monitoring, because the highly standardized reporting and additional regulatory monitoring in these industries limits clients' ability and motivation to differentiate themselves on disclosure quality dimensions. Accordingly, the ability of audit firms to add value via disclosure quality is limited in regulated industries.[2] The high level of disclosure required in regulated industries also reduces the cost of information transfer to competitors; therefore, the signaling role of the audit firm is also diminished in regulated industries.

We utilize standard measures of disclosure quality and audit firm industry specialization in our tests for an association between the two constructs. We use analysts' disclosure quality evaluations reported in the annual AIMR Corporate Information Committee Reports as a proxy for disclosure quality (Lang and Lundholm, 1993, 1996; Sengupta, 1998). We use the proportion of two-digit SIC industry sales audited by each audit firm as a proxy for auditor industry specialization (Palmrose, 1986; Mayhew and Wilkins, 2003). The Big Six audit all of our sample firms, so the industry specialization measures do not reflect a dichotomy between the Big Six and other accounting firms.

We test our hypotheses with cross-sectional analyzes employing control variables identified by Lang and Lundholm (1993). We document a positive association between industry-specialist audit firms and the AIMR overall disclosure quality ranking as well as the annual report rankings. Consistent with our expectations, a partition of the data into regulated and unregulated industries suggests that a strong positive association between audit firm industry specialization and disclosure quality in unregulated industries drives our overall results. There does not appear to be any such association in regulated industries. An examination of AIMR Disclosure Quality Awards given by the same analyst committees to the top firms in each industry further corroborates our conclusions. Sensitivity tests for a number of alternative measures of audit firm industry specialization support our findings and suggest that individual Big Six audit firms do not drive the results.

We present our theory of audit firm industry specialization and our hypotheses in the next section. Section 2 provides a discussion of the empirical proxies we employ. Section 3 presents the research design and data. Section 4 describes the results and presents tests using alternative measures of audit firm industry specialization. The final section summarizes our findings.

1. Theory and Hypotheses

We start with a basic description of the auditor's motivation to specialize along industry lines and why clients demand industry-specialist firms. We then discuss why all clients do not demand specialist auditors and why auditors limit their total exposure to a given industry. This discussion leads directly to our hypotheses about the association between industry specialization and disclosure quality.

1.1. Why Do Auditors Specialize?

One goal of an audit firm is to identify ways to differentiate itself from competitors in serving client needs. Differentiation enables the auditor to compete on dimensions other than price to win and retain clients. We start with the assumption that all audit clients have a unique set of characteristics, and that audit firms must adjust to client characteristics to meet client needs (Chan et al., 2001). This creates incentives for audit firms to develop specializations to meet these client needs in ways that competitors cannot easily duplicate. We focus on industry membership as an important dimension with which audit firms align themselves with specific client characteristics and related demands.[3] Such specialization is a particularly valuable dimension on which to differentiate because it allows the audit firm to use its differentiation strategy to service a relatively large group of clients with similar characteristics.

There is evidence that within individual industries, the Big Six firms with the largest market shares have expanded their market shares over the last 20 years (Hogan and Jeter, 1999). This suggests that audit firms are meeting client needs by

supplying industry specific value to their clients. It also suggests that market share is a reasonable way to measure the industry specialization of the audit firm.

1.2. Why Do Clients Demand Specialization?

We assert that clients select auditors as part of their overall disclosure strategy. Clients demand industry specialized auditors for (1) the value added benefits provided by an industry-specialist auditor, including potentially lower fees, enhanced audit quality and disclosure advice and (2) as a signaling mechanism to investors that the client intends to provide enhanced disclosures.

Industry-specialist auditors benefit clients in a number of ways. First, prior research suggests that as an audit firm's level of specialization, as measured by market share, increases, the audit firm obtains greater economies of scale (Danos and Eichenseher, 1982). Mayhew and Wilkins (2003) show that in general, competition in the audit market results in audit firms sharing this cost advantage with clients; however, clients with market shares that greatly exceed their nearest competitors earn significant fee premiums.

Second, industry-specialist auditors gain more industry specific knowledge than non-specialists. Recent evidence suggests that industry-specialists are able to use this knowledge to provide more effective audits as evidenced by higher earnings quality (Balsam et al., 2003; Gramling et al., 2000). Industry-specialists have more industry expertise that enables them to identify misstatements more effectively. Their expertise comes from serving other clients in the same industry and learning and sharing best practices across the industry. Industry-specialists also have more incentive to correct or report identified misstatements to protect their market shares. The increase in audit quality should also impact disclosure quality by enhancing financial statement credibility.

Finally, our main theory is that industry-specialist auditors use their industry specific knowledge to assist clients in developing and disseminating enhanced disclosures. In support of this view, the financial press suggests that clients typically seek auditors "who understand their industries" (Goff, 2002). Baruch Lev (2002) recently argued for the break-up of Andersen into "three or four new accounting firms structured along industry/technology lines" as an alternative solution to the collapse of Andersen. His argument was based on the belief that smaller industry-specialist firms could compete with the remaining big four firms by providing industry "specialized procedures and know-how." Behn et al. (1997) also show that industry specialization is a key determinant of client satisfaction and that clients highly value auditor advice beyond basic GAAP.

We are aware of at least two specific examples of firms assisting clients in enhancing industry specific disclosures. First, Coopers and Lybrand LLP (now PriceWaterhouseCoopers LLP (PWC)) provides its clients with an annual assessment of important disclosure issues for specific industries (Coopers & Lybrand, LLP, 1998). This annual assessment provides industry specific guidance on hot issues with investors and analysts, including potential questions that may

arise at the annual shareholder meeting. Second, a group of PWC partners has written a book on enhanced financial reporting (Eccles et al., 2001), which discusses developing industry and business specific value measuring metrics—both financial and nonfinancial—and ultimately disclosing these metrics to the public. In discussion with one of the authors, we learned that PWC works with a number of clients in target industries to implement these enhanced disclosure practices with the ultimate goal of developing more transparency in clients' corporate reporting.

To provide further support for industry-specialists providing more services beyond the basic audit than non-specialists, we utilize the new fee disclosures recently required by the SEC for US companies, which clearly differentiate between fees charged for basic audit services and fees for non-audit services. A comparison of the ratio of non-audit to total fees suggests that industry-specialists provide significantly more non-audit services than non-specialists.[4] This is consistent with industry-specialists providing more services beyond the basic audit than non-specialists.

1.3. Why Don't All Clients Demand Specialization?

In addition to providing clients with value-added benefits, industry specialist auditors aid clients in signaling their intention to provide high quality disclosures. The signaling aspect of industry specialization also explains why not all clients want to hire an industry specialist. While clients benefit from improved disclosure quality, they also bear costs related to improved disclosures.[5] For some clients the costs of disclosure quality outweigh the benefits. In the context of hiring a specialist auditor, client costs could be substantial. Not only would a client who bears competitive costs of disclosure (i.e., trade secrets or strategic advantages) want to avoid enhanced disclosures in general, it would also want to avoid an auditor who services a large number of its competitors. Auditors develop intimate knowledge of client business practices. This knowledge can potentially be transferred to other audit clients in the same industry. The risk of information transfer to competitors increases with the audit firm's industry market share. In support of this information transfer concern, prior research suggests that clients in some industries try to avoid hiring the same auditor as their major competitors to protect against the transfer of proprietary information (Kwon, 1996). For example, when Ernst and Whinney and Arthur Young merged in 1989 into Ernst and Young, the merged firm temporarily became the auditors of both Coca Cola and Pepsi. Both clients found this arrangement unacceptable. As a result, Pepsi hired a new auditor (Kwon, 1996).

Given the potential costs of information transfer, hiring an industry-specialist auditor becomes a credible signal of a client's choice to provide enhanced disclosures to the market. A client trying to mimic the signal inherent in hiring an industry-specialist subjects auditor itself to potential costly information transfer to competitors. In addition, the client is subject to a higher quality audit that could result in the auditor discovering and reporting information that the client would like

to avoid reporting to the market. These costs are in addition to any additional fees charged by industry specialists.[6]

Audit firms themselves also may want to avoid becoming overly concentrated in one client industry. The collapse of the Savings and Loan industry in the late 1980s suggests that audit firms may want to remain somewhat diversified to better balance risk in their overall client portfolio. Simunic and Stien (1990) suggest that audit firms must trade-off the benefits of specialization and the loss of portfolio diversification.

1.4. Hypotheses

The above discussion links audit firm industry specialization with higher client disclosure quality. An industry-specialist audit firm possesses more industry specific knowledge and expertise in aiding clients with disclosure related issues. The specialist audit firm can, in most cases, provide this knowledge to clients more cost effectively than clients can independently acquire this knowledge. The hiring of a specialist also serves as a credible signal of an intention to provide high quality disclosures because firms that try to mimic the strategy will suffer from information transfer (a) to competitors because specialists audit more competitors than non-specialists, and/or (b) to the market because specialists provide higher quality audits and will uncover any deficiencies in client reporting. This leads to our first hypothesis stated in alternative form.

H1: Disclosure quality is higher for clients employing industry-specialist audit firms than for clients employing non-specialist audit firms.

We also consider whether the demand for specialist auditors is homogeneous across industries. Specifically, differences in the level of information asymmetry and monitoring by other parties may diminish the importance of enhanced disclosure and accordingly the value of a specialist auditor. All firms in our sample file financial statements with the SEC. The SEC provides a minimum level of financial statement monitoring that we assume is homogeneous across all the industries and firms in our sample.[7]

In addition to SEC monitoring, government regulation in certain industries establishes an agency overseeing the economic regulation of the industry, serving a monitoring role similar to the role of the auditor in the financial reporting process. These agencies typically specify reporting requirements beyond basic financial statements for firms under their jurisdiction. They also monitor compliance with the reporting requirements. In some cases (e.g., banking and insurance), these monitoring agencies even conduct their own audit investigations. While audit firms may specialize in their ability to assist clients in fulfilling these specialized reporting requirements, an audit firm's value in enhancing disclosure and providing a signal of disclosure quality is diminished by the reduction in information asymmetry generated by the additional regulatory oversight.[8] Given the already substantial levels of disclosure in economically regulated industries, the client has little reason to

expand disclosure. In fact, the client may actually want to limit enhanced disclosure to avoid additional regulatory scrutiny. Accordingly, the signaling value of the industry-specialist is also reduced because there is little information not already in the public domain due to the regulatory disclosures, therefore clients are not concerned about the cost of additional information being revealed and transferred by a specialist.

To compare the relation between disclosure quality and audit firm industry specialization across regulated and unregulated industries, we conduct a second set of analyses to test the following hypothesis.

H2: Disclosure quality is higher (the same) for clients employing industry-specialist audit firms than for clients employing non-specialist audit firms in unregulated (regulated) industries.

2. Empirical Proxies

In this section, we describe the empirical proxies that represent our main theoretical constructs—audit firm industry specialization and client disclosure quality.

2.1. Industry Specialization Proxy

We define industry specialization as an increasing function of market share. We measure market share using the total sales audited by an audit firm within an industry (Palmrose, 1986). We define an industry as all companies within each two-digit primary Standard Industry Classification (SIC) code in the Compustat database. Prior research shows a correlation between audit fees and client size measured by sales or assets (Simunic, 1980; Palmrose, 1986). Consistent with DeAngelo's (1981) argument that audit firm size correlates with audit quality, our definition of specialization assumes that audit firms with large market shares, and therefore industry specific audit fees, have strong incentives to deliver high quality audits and services.

We report results based on both a continuous ranking of the industry sales audited by the audit firms and a 20% market share cut-off.[9] We recalculate the audit firm specialization measure for each of the years in our sample. Prior to the consolidation of the Big Eight into the Big Six in 1989, researchers generally defined an audit firm as an industry-specialist if it audits more than 10% of firms, fees or sales in an industry (Craswell et al., 1995; DeFond, 1992). After the consolidation, a Big Six accounting firm has more than an equal share of an industry if it audits 16% or more of the industry; therefore, we set our measure of industry specialization at 20% of industry sales audited. A 20% cut-off insures that in every industry the entire Big Six cannot be classified as specialists.

Krishnan (2001) argues that market share may not always signal quality. She notes that specialization evolves to serve market niches, and that these niches need not be very large. Our robustness checks consider the possibility of small market share specialization by partitioning audit firm specialization into less than 10%, 10–20%, and more than 20% of market share. We also consider a within-firm measure of industry specialization, based on the sales an audit firm audits within an industry as a portion of the total sales it audits over all industries.

2.2. Disclosure Quality

Our measure of disclosure quality is based on analysts' rankings of company disclosures. Each year in our sample, the AIMR Corporate Information Committee selects buy and sell-side analysts to form industry-specific committees that evaluate the disclosure quality of firms in target industries. Each industry subcommittee determines its own criteria and scoring system and then selects the firms to evaluate. Subcommittees evaluate the adequacy of disclosure for three categories: annual published information, quarterly and other published information, and investor relations. Within the categories, each industry subcommittee identifies the important aspects of disclosure for that industry and then assigns a score to each firm. The subcommittees then weight each category to arrive at an overall score. While industry subcommittees differ with respect to what they discuss in their reports and the scoring system they employ, analysts' concern rests with the timeliness and completeness of firm disclosures.[10] A majority of subcommittees report scores in each of the three categories and an overall score. However, some industries report only an overall score, some only rank the firms, and a minority provides only qualitative evaluations. Finally, the AIMR subcommittees select up to two firms from each industry for recognition of excellence in corporate reporting.

Our main interest lies in the impact of industry-specialist audit firms on disclosure quality. Lang and Lundholm (1993) infer that the main driver of the AIMR scores is the informativeness of client disclosures. We conjecture that specialist audit firms help clients improve the informativeness of disclosures. The auditors' biggest impact on disclosure quality likely occurs in the annual report, and to a lesser extent in the quarterly and other published information scores. However, not all industry subcommittees provide scores for these categories. Because the overall disclosure quality scores and rankings implicitly capture financial reporting quality, and most AIMR industry committees provide overall scores or rankings, we use the overall measures for our main analyzes.[11] We provide additional analyses using the annual report scores for those industries that supply it. We also use the Award for Excellence/Letter of Commendations as a further test of the impact of industry-specialization on disclosure quality.

3. Research Design and Data

3.1. Research Design

We use the same rank regression methods as Lang and Lundholm (1993). We rank dependent and independent variables within each AIMR industry and then convert to fractions: (rank-1)/(number of firms—1).[12] The conversion yields the fraction of a firm's rank in the industry, so that the highest rank firm receives a one and the lowest receives a zero. The rank regressions allow us to pool data cross-sectionally even though separate analyst committees evaluate the firms using different scoring systems. We do not explore changes in disclosure quality related to a change in audit firm because our sample includes only a small number of auditor switches. To reduce the risk of incorrect inferences from pooled time-series data, we conduct tests on average regression coefficients from year-by-year regressions using the same method as Fama and MacBeth (1973).[13]

3.2. Control Variables

The choice of an industry specialized auditor may be related to other firm characteristics that contribute to disclosure quality. We include Lang and Lundholm's variables in our multivariate analyzes to control for potential alternative explanations to our hypotheses. Lang and Lundholm (1993) examine the relation between AIMR scores and firm characteristics over the 1985–1989 period. They provide evidence that AIMR scores increase with firm size and firm performance measured by current returns. Firms issuing securities in the current and subsequent two periods also receive higher AIMR scores. Conversely, Lang and Lundholm's (1993) evidence suggests that analysts award lower scores for firms with high earnings/returns correlations. They provide weak evidence of a negative association of the volatility of annual returns with AIMR scores.

We test H1 and H2 using the Specialist variable in the following model:

$$\begin{aligned}\text{AIMR score}_i = {} & \alpha_1 + \beta_1 \text{Market value}_i + \beta_2 \text{Earnings-return correlation}_i + \\ & \beta_3 \text{Standard deviation of returns}_i + \beta_4 \text{Forecast error}_i + \\ & \beta_5 \text{Return}_i + \beta_6 \text{Offer}_i + \beta_7 \text{Specialist}_i. \end{aligned} \quad (1)$$

AIMR score_i = The sample firm's AIMR rank for overall (annual) disclosure quality within its industry. We hand collect this data from the AIMR Corporate Information Committee Reports. AIMR labels the reports 1990–1991, 1991–1992, etc. which we categorize as 1990, 1991, etc.

Market value_i = The market value of outstanding equity at the beginning of the fiscal year—Compustat Data Item 25 multiplied by Compustat Data Item 24.

Earnings-return correlation$_i$ = The correlation between annual stock returns from the CRSP Monthly Stock File and annual earnings in the Compustat Annual Tapes, Data Item 58, for the 10 years prior to the current year. We also include firms with fewer than 10 years of data, as long as they have at least four years of earnings and returns information.

Standard deviation of returns$_i$ = The standard deviation of annual market-adjusted stock returns for the 10 years prior to the current fiscal year. We base the market-adjusted returns on the difference between annual firm returns and the annual equal-weighted market returns, both from the CRSP Monthly Stock File. We also include firms with fewer than 10 years of data, as long as they have at least four years of earnings and returns information.

Forecast error$_i$ = I/B/E/S actual earning per share less the I/B/E/S mean consensus forecast earning per share at the beginning of the fiscal year, divided by the I/B/E/S reported price per share at the beginning of the fiscal year.

Return$_i$ = The annual firm return less the annual equal-weighted market return for the fiscal year from the CRSP Monthly Stock file.

Offer$_i$ = An indicator variable equal to one if the firm files a debt or equity registration statement in the current fiscal year or in the next two fiscal years, and zero otherwise. We hand collect the data from the Capital Changes Reporter and Investment Dealer's Digest.

Specialist$_i$ = Industry-specialist as defined by Auditor Industry Share and Auditor 20% Industry Share. Auditor Industry Share is the percentage of sales the client's audit firm audits in the client's two-digit SIC code as reported by Compustat.[14] Auditor 20% Industry Share is an indicator variable equal to one if the client's audit firm audits at least 20% of the sales in the client's two-digit SIC code as reported by Compustat, and zero otherwise. Auditor 20% industry share for financial companies equals one if the client's audit firm audits at least 20% of the sales in the client's industry for companies in our sample.

We test H1 on the relationship between audit firm industry specialization and the overall disclosure quality score by using the rank fraction of the overall disclosure quality score as the dependent variable. We then test H1 using the rank fraction of the annual published information score as the dependent variable. The auditors' close affiliation with the annual report suggests the annual report score should capture the influence of the auditor on disclosure quality.

As an additional test, we replace the dependent variable in the above regression with an indicator variable (Award) that captures whether firms receive an Award for Excellence or Letter of Commendation in Corporate Reporting from the AIMR

committees. The resulting logistic model is then estimated using the same control and treatment variables used in the basic regression model.

We also use our regression model to test H2 on the association between disclosure quality and audit firm specialization in sub-samples of regulated and unregulated industries. Prior studies have typically taken a broad view of regulation encompassing a wide range of social welfare protections. We partition regulated and unregulated industries to capture differences in the level of monitoring related to financial reporting or performance. We partition industries based on economic regulation but not environmental, safety or health regulation. We define regulated industries as industries under the heading "Economic Regulation" in Table 1 of Weiss and Klass (1986).[15] Hogan and Jeter (1999) use the same table to partition regulated industries, but unlike our partition they also include industries under the heading "Environmental, Safety and Health Regulation" in their regulated sample.[16]

While the ability of regulated firms to differentiate themselves on disclosure quality may be constrained, they can still differentiate themselves on other AIMR categories—quarterly and other published information, and investor relations. Firms may also exhibit differences in terms of perceived disclosure quality based on performance or structural characteristics similar to those discussed in Lang and Lundholm (1993) and used by us as control variables.

While we expect our control variables to apply to both regulated and unregulated industries, the relative importance of some control variables may differ across regulated and unregulated industries. Nwaeze (2000) documents an asymmetric stock price response to earnings surprises in regulated industries not present in unregulated industries. The basic intuition for his results is that in rate-regulated industries a regulated firm's failure to meet earnings targets may be good news, since it can then argue for a rate increase and rate regulated industries may not have to "give back" positive earnings surprises. Lang and Lundholm (1993) document a relation between disclosure quality ratings and both stock price reactions to earnings (i.e., earnings return correlation) and forecast errors. Accordingly, Nwaeze's findings suggest that earnings-return correlation and forecast errors may not have the same impact on perceived disclosure quality across regulated and unregulated industries. We consider this possibility by examining differences in the importance of our control variables on disclosure quality between regulated and unregulated industries.

3.3. Data

We examine the relation between the AIMR scores and audit firm specialization for six AIMR Corporate Information Committee annual reports issued from 1990–1991 to 1995–1996. The sample intentionally omits data prior to the 1989 consolidation of the Big Eight into the Big Six. By employing only post-consolidation data, we avoid any shifts in industry specialization associated with the consolidation and provide a more valid test of association.

Our sample starts with all firms the analyst subcommittees include in the AIMR reports. A small number of industries provide only qualitative evaluation

disclosures, so we omit them from the analyzes. We use the firm-years that include all the data necessary to calculate our control variables on Compustat, CRSP, and I/B/E/S databases from the remaining AIMR industries.[17]

Table 1 lists the descriptive statistics for the overall sample (Panel A) and for subsamples of unregulated (Panel B) and regulated (Panel C) industries. We report unadjusted (i.e., non-ranked) numbers in the descriptive statistics with the exception of the overall and annual report scores, which we report as ranked fractions. In our

Table 1. Descriptive statistics.

			Specialist		Non-Specialist		
Variable	*N*	Mean	*N*	Mean	*N*	Mean	*t*-statistic
Panel A: All Observations							
Overall (rank) score	1,998	0.50	897	0.52	1,101	0.47	2.49*
Annual (rank) score	1,351	0.50	634	0.52	717	0.48	2.81**
Award	1,998	0.08	897	0.11	1,101	0.06	4.19**
Market value	1,932	4,966	878	5,494	1,054	4,525	2.23*
Earnings-return correlation	1,712	0.18	803	0.20	909	0.15	3.17**
Standard deviation returns	1,729	0.27	806	0.26	923	0.28	2.63**
Forecast error	1,877	−0.04	854	−0.03	1,023	−0.05	1.10
Return	1,901	0.01	863	−0.01	1,038	0.03	2.35*
Offer	1,998	0.47	897	0.48	1,101	0.47	0.80
Auditor Industry Share	1,998	0.21	897	0.31	1,101	0.12	44.33**
Panel B: Unregulated Industries							
Overall (rank) score	1,333	0.50	544	0.54	789	0.47	3.62**
Annual (rank) score	1,095	0.50	494	0.53	601	0.47	3.41**
Award	1,333	0.08	544	0.11	789	0.05	4.13**
Market value	1,301	5,894	533	7,224	768	4,971	3.42**
Earnings-return correlation	1,124	0.16	477	0.17	647	0.14	1.61
Standard deviation returns	1,132	0.28	477	0.27	655	0.28	1.88†
Forecast error	1,246	−0.02	513	−0.02	733	−0.02	0.30
Return	1,271	0.01	522	−0.02	749	0.04	3.03**
Offer	1,333	0.46	544	0.48	789	0.45	1.08
Auditor Industry Share	1,333	0.20	544	0.33	789	0.12	33.80**
Panel C: Regulated Industries							
Overall (rank) score	665	0.50	353	0.49	312	0.51	0.70
Annual (rank) score	256	0.50	140	0.49	116	0.51	0.40
Award	665	0.08	353	0.09	312	0.06	1.16
Market value	631	3,051	345	2,821	286	3,327	1.53
Earnings-return correlation	588	0.22	326	0.25	262	0.18	2.46*
Standard deviation returns	597	0.24	329	0.24	268	0.25	0.56
Forecast error	631	−0.08	340	−0.04	290	−0.12	1.37
Return	630	0.02	341	0.02	289	0.01	0.05
Offer	665	0.51	353	0.50	312	0.52	0.52
Auditor Industry Share	665	0.22	353	0.29	312	0.13	22.31**

Data for all firms covered by AIMR Information Committee reports from 1990–1991 to 1995–1996 with necessary data available on Compustat, IBES and CRISP. Regulated industries include all industries under the heading "Economic Regulation" in Table 1 of Weiss and Klass (1986) plus insurance and aerospace. For AIMR industries that include more than one two-digit SIC code, we categorize the AIMR industry based on the regulated/unregulated status of the majority of the firms within the AIMR industry. Specialist audit firms audit more than 20% of the total sales within a two-digit SIC code. Overall (rank) score is the overall AIMR disclosure quality score as reported in the AIMR Corporate Information Committee and ranked within each AIMR industry and converted to fractions: (rank-1)/(number of firms—1). Annual (rank) score is the annual report AIMR disclosure quality score as reported in the AIMR Corporate Information Committee Reports and ranked within each AIMR industry and converted to fractions: (rank-1)/(number of firms—1). Award is equal to one if the client receives an Award for Excellence or Letter of Commendation in Corporate Reporting from the AIMR committees and zero otherwise. Market value is the beginning of the fiscal year market value of outstanding equity in millions of dollars. Earnings-return correlation is the correlation between annual returns and annual earnings for the preceding ten years. It includes firms as long as they have at least four years of earnings and returns information. Standard deviation returns is the standard deviation of annual market-adjusted stock returns calculated over the 10 years prior to the current fiscal year. It also includes firms with at least four years of data. Forecast error is the I/B/E/S actual earning per share less the I/B/E/S mean consensus forecast earning per share at the beginning of the fiscal year, divided by the I/B/E/S reported price per share at the beginning of the fiscal year. Return is annual firm return less the annual equal-weighted market return. Offer is an indicator variable that equals one if the firm filed a debt or equity registration statement in the current fiscal year or next two fiscal years, and zero otherwise. Auditor Industry Share is the percentage of sales the firm's audit firm audits in the firm's two-digit SIC code. Auditor Industry Share ratios for non-financial industries are from Compustat. Compustat does not include audit firm codes for financial industries. For financial industries we base audit firm shares on the companies in our sample. †, *, and ** indicate significance at 0.10, 0.05 and 0.01 for two-tailed test.

regression analyses, we convert the variables into rank fractions by industry as discussed previously.

Not surprisingly, given our focus on firms followed by analysts, the typical firm in our sample has a large market capitalization (i.e., almost $5 billion). The median audit firm in the sample audits 19% of the firms in each two-digit SIC code based on industry sales. This supports our use of a 20% cut-off for defining industry specialization. Regulated industries appear to have higher concentrations of audit firms consistent with prior research (Hogan and Jeter, 1999). The descriptive statistics for the sub-samples suggest some structural differences between the unregulated and regulated sub-samples. Market values are lower on average and do not vary as much for regulated industries. Earnings-return correlation is stronger in the regulated sample than the unregulated sample, which is consistent with Nwaeze (2000). Standard deviation of returns, forecast error, returns, and offer are similar in terms of means and distributions across the two samples.

Univariate *t*-tests comparing the samples split into specialist and non-specialist partitions suggests that firms that select specialists are structurally different than firms that select non-specialists. However, when we break the sample down into unregulated (Table 1, Panel B) and regulated (Table 1, Panel C), it becomes clear that only in unregulated industries are there signficant differences in firms that select specialists or non-specialists. The univariate tests suggest that our control variables are necessary to control for differences in firms that hire specialists.

4. Results

Table 1 provides univariate evidence on our hypotheses. We use 20% of sales as the cut-off for all univariate specialists versus non-specialists comparisons. Consistent with H1, we see that in the overall sample, firms that hire industry-specialist auditors have higher overall and annual disclosure quality scores and are more likely to receive a disclosure award. Panels B and C show that the higher scores are only present in unregulated industries as predicted in H2. These results are only suggestive, as it is clear that we need to control for other factors in evaluating the association between disclosure quality and auditor industry specialization.

Table 2 provides correlation analyzes of all of the variables included in our regression analyzes. Again, the correlation between disclosure quality metrics and auditor industry specialization metrics supports H1. There is also significant correlation between market value and our industry specialization metrics. This association is expected as our industry specialization measure is influenced by the size of firms within industries. We provide sensitivity tests later to assess the influence of size on our analysis.

Table 3 summarizes the regression results with the overall score (Panel A) and annual report score (Panel B) as the dependent variable for all industries. The all industries model shows a positive association between audit firm industry specialization and overall (annual report) disclosure quality in support of H1.[18]

Table 4 shows that when we break the sample into regulated (Panels A and B) and unregulated (Panels C and D) industries, the results are driven entirely by unregulated industries, consistent with H2. Audit firm industry specialization is significantly related to disclosure quality in the unregulated industry sample, but not related to disclosure quality in the regulated industry sample for both measures of specialization.

The average coefficients on our control variables differ slightly across the two sub samples. The largest difference is that forecast error explains a significant portion of variation in disclosure quality in the regulated sample but not the unregulated sample. We also note that earnings-return correlation explains more variation in scores in the unregulated sample. Nwaeze (2000) shows an asymmetric stock price reaction to earnings surprises (a metric similar to our forecast error measure) for regulated companies but not for a sample of manufacturing companies. To the extent that analysts' disclosure quality scores reflect structural relations between stock price and earnings, as conjectured by Lang and Lundholm (1993), the differences we document across regulated and unregulated industries are not surprising. Given this explanation, we do not think the differences in the association between forecast error or the earnings-return correlation and disclosure quality are related to our conjectures of an association between our industry specialization measure and disclosure quality. Accordingly, we do not expect it to impact our predictions about audit firm industry specialization.

We also examine Awards of Excellence awarded by each analyst industry committee for evidence of an association between disclosure quality and audit firm industry specialization. Table 5 shows the results of average year-over-year coefficients from our logistic model. The table suggests industry specialization is

Table 2. Correlation for variables used in the regression analysis.
Pearson Correlation Coefficients above the diagonal and Spearman Correlation coefficients below the diagonal, *t*-statistics, and number of observations used in the calculations. We rank all continuous independent and dependent variables within each AIMR industry and then convert to fractions: (rank-1)/(number of firms—1).

	Overall	Annual	Award	Market Value	Earnings-Return Correlation	Standard Deviation Returns	Forecast Error	Return	Offer	Auditor 20% Industry Share	Auditor Industry Share
Overall	1	0.808	0.429	0.174	0.061	−0.026	0.122	0.085	0.076	0.053	0.086
		<0.001	<0.001	<0.001	0.012	0.288	<0.001	<0.001	0.001	0.018	<0.001
	1,998	1,351	1,998	1,932	1,712	1,729	1,877	1,901	1,998	1,998	1,998
Annual	0.809	1	0.398	0.209	0.039	−0.102	0.150	0.065	0.083	0.075	0.052
	<0.001		<0.001	<0.001	0.184	<0.001	<0.001	0.020	0.002	0.006	0.054
	1,351	1,351	1,351	1,315	1,189	1,201	1,273	1,294	1,351	1,351	1,351
Award	0.429	0.398	1	0.114	0.060	0.002	0.048	0.049	0.086	0.093	0.053
	<0.001	<0.001		<0.001	0.013	0.945	0.036	0.031	<0.001	<0.001	0.017
	1,998	1,351	1,998	1,932	1,712	1,729	1,877	1,901	1,998	1,998	1,998
Market value	0.174	0.209	0.113	1	−0.025	−0.152	0.178	0.143	0.105	0.112	0.096
	<0.001	<0.001	<0.001		0.311	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
	1,932	1,315	1,932	1,932	1,684	1,688	1,837	1,869	1,932	1,932	1,932
Earnings-return	0.059	0.040	0.057	−0.022	1	0.185	0.018	0.052	0.103	0.047	0.052
correlation	0.015	0.168	0.019	0.357		<0.001	0.456	0.033	<0.001	0.054	0.032
	1,712	1,189	1,712	1,684	1,712	1,712	1,636	1,700	1,712	1,712	1,712
Standard	−0.025	−0.101	−0.003	−0.147	0.177	1	0.031	0.056	0.049	0.030	0.033
deviation	0.299	0.001	0.912	<0.001	<0.001		0.207	0.019	0.044	0.217	0.172
returns	1,729	1,201	1,729	1,688	1,712	1,729	1,651	1,714	1,729	1,729	1,729
Forecast	0.123	0.152	0.049	0.181	0.019	0.032	1	0.446	0.009	−0.035	0.026
error	<0.001	<0.001	0.035	<0.001	0.448	0.189		<0.001	0.705	0.127	0.260
	1,877	1,273	1,877	1,837	1,636	1,651	1,877	1,821	1,877	1,877	1,877
Return	0.086	0.066	0.049	0.143	0.050	0.056	0.448	1	0.031	−0.033	0.011
	0.000	0.017	0.034	<0.001	0.038	0.021	<0.001		0.172	0.146	0.620
	1,901	1,294	1,901	1,869	1,700	1,714	1,821	1,901	1,901	1,901	1,901
Offer	0.076	0.083	0.086	0.105	0.103	0.045	0.009	0.031	1	0.0178	0.0203
	0.001	0.002	<0.001	<0.001	<0.001	0.060	0.713	0.173		0.4258	0.3641
	1,998	1,351	1,998	1,932	1,712	1,729	1,877	1,901	1,998	1,998	1,998
Auditor 20%	0.054	0.075	0.093	0.112	0.044	0.033	−0.035	−0.033	0.018	1	0.7036
Industry Share	0.016	0.006	<0.001	<0.001	0.067	0.175	0.128	0.148	0.426		<0.001
	1,998	1,351	1,998	1,932	1,712	1,729	1,877	1,901	1,998	1,998	1,998
Auditor Industry Share	0.087	0.051	0.052	0.096	0.046	0.031	−0.025	−0.011	−0.018	0.701	1
	<0.001	0.061	0.020	<0.001	0.056	0.199	0.288	0.643	0.426	<0.001	
	1,998	1,351	1,998	1,932	1,712	1,729	1,877	1,901	1,998	1,998	1998

For definitions, see notes to Table 1. Auditor 20% Industry Share is one if the firm's audit firm audits 20% of more of sales in the client's two-digit SIC code and zero otherwise.

Table 3. Regression analysis of AIMR scores.
Average regression coefficients from year-by-year rank regressions, where we rank all continuous independent and dependent variables within each AIMR industry and then convert to fractions: (rank-1)/(number of firms—1). We estimate annual rank regressions for each AIMR Corporate Information Committee report from 1990–1991 through 1995–1996 (six years). Reported *t*-statistics test whether the average regression coefficient equals zero.

$$\text{AIMR score}_i = \alpha_1 + \beta_1 \text{ Market value}_i + \beta_2 \text{ Earnings-return correlation}_i + \beta_3 \text{ Standard deviation returns}_i + \beta_4 \text{ Forecast error}_i + \beta_5 \text{ Return}_i + \beta_6 \text{ Offer}_i + \beta_7 \text{ Specialist}_i.$$

	Average Regression Coefficient	*t*-statistic	Average Regression Coefficient	*t*-statistic
Panel A: AIMR Overall Score—All Industries				
Intercept	0.30	8.14**	0.29	8.44**
Market value	0.22	12.27**	0.22	11.23**
Earnings-return correlation	0.06	3.35*	0.05	3.22*
Standard deviation returns	− 0.05	− 1.09	− 0.05	− 1.11
Forecast error	0.07	2.45†	0.07	2.57*
Return	0.01	0.15	0.00	0.07
Offer	0.03	3.53*	0.03	4.24**
Auditor 20% Industry Share	0.05	3.20*		
Auditor Industry Share			0.08	4.50**
Average adjusted R^2	9.0%	6.63**	8.8%	7.67**
Panel B: AIMR Annual Report Score—All Industries				
Intercept	0.32	6.94**	0.32	6.74**
Market value	0.26	7.49**	0.26	6.75**
Earnings-return correlation	0.03	1.84	0.03	1.78
Standard deviation returns	− 0.09	− 1.62	− 0.09	− 1.62
Forecast error	0.09	1.38	0.09	1.37
Return	− 0.02	− 0.36	− 0.02	− 0.43
Offer	0.03	2.51†	0.03	2.30†
Auditor 20% Industry Share	0.05	4.78**		
Auditor Industry Share			0.05	2.73*
Average adjusted R^2	12.8%	5.14**	12.5%	5.49**

Terms are defined in notes to Tables 1 and 2.

associated with disclosure quality. Once again, when we conduct our analysis separately for unregulated (Panel B) and regulated (Panel C) industries, we see unregulated industries drive the overall results, consistent with H2.

4.1. Robustness Tests

To control for possible individual Big Six firm effects, we add individual indicator variables for each of the Big Six. Our analyses indicate that one or more of the

Table 4. Average regression coefficients from year-by-year rank regression.
Regression analysis of AIMR score for regulated and unregulated industries. We rank all continuous independent and dependent variables within each AIMR industry and then convert to fractions: (rank-1)/(number of firms — 1). We estimate Annual rank regressions for each AIMR Corporate Information Committee report from 1990–1991 through 1995–1996 (six years). Reported *t*-statistics test whether the average regression coefficient equals zero.

$$\text{AIMR score}_i = \alpha_1 + \beta_1 \text{ Market value}_i + \beta_2 \text{ Earnings-return correlation}_i + \beta_3 \text{ Standard deviation returns}_i + \beta_4 \text{ Forecast error}_i + \beta_5 \text{ Return}_i + \beta_6 \text{ Offer}_i + \beta_7 \text{ Specialist}_i.$$

	Average Regression Coefficient	*t*-statistic	Average Regression Coefficient	*t*-statistic
Panel A: Regulated Industries—Overall AIMR Score				
Intercept	0.28	2.84*	0.28	2.82*
Market value	0.19	3.55*	0.19	3.55*
Earnings-return correlation	0.04	0.71	0.04	0.69
Standard deviation returns	− 0.05	− 0.72	− 0.05	− 0.77
Forecast error	0.12	3.55*	0.13	3.35*
Return	0.02	0.22	0.01	0.14
Offer	0.06	2.07†	0.07	2.11†
Auditor 20% Industry Share	0.02	0.47		
Auditor Industry Share			0.99	0.37
Average adjusted R^2	9.1%	3.49*	8.5%	3.61*
Panel B: Regulated Industries—Annual Report AIMR Score				
Intercept	0.25	1.77	0.33	2.87*
Market value	0.24	2.84*	0.24	2.78*
Earnings-return correlation	− 0.04	− 0.63	− 0.04	− 0.60
Standard deviation returns	− 0.11	− 0.95	− 0.13	− 1.36
Forecast error	0.26	3.76*	0.29	5.42**
Return	− 0.09	− 0.94	− 0.11	− 1.09
Offer	0.14	2.05†	0.11	2.40†
Auditor 20% Industry Share	0.04	0.51		
Auditor Industry Share			− 0.02	− 0.29
Average adjusted R^2	25.7%	2.36†	23.1%	2.97*
Panel C: Unregulated Industries—Overall AIMR Score				
Intercept	0.30	5.56**	0.28	5.39**
Market value	0.25	41.00**	0.25	25.61**
Earnings-return correlation	0.07	2.26†	0.07	2.22†
Standard deviation returns	− 0.05	− 0.87	− 0.05	− 0.88
Forecast error	0.03	0.54	0.04	0.59
Return	0.03	0.50	0.02	0.43
Offer	0.03	1.95	0.03	1.76
Auditor 20% Industry Share	0.07	7.85**		
Auditor Industry Share			0.11	6.10**
Average adjusted R^2	11.3%	6.53**	11.3%	7.08**

Table 4. Continued.

	Average Regression Coefficient	*t*-statistic	Average Regression Coefficient	*t*-statistic
Panel D: Unregulated Industries—Annual Report AIMR Score				
Intercept	0.32	6.29**	0.31	6.06**
Market value	0.27	10.85**	0.27	9.82**
Earnings-return correlation	0.05	1.77	0.04	1.70
Standard deviation returns	−0.06	−1.28	−0.06	−1.24
Forecast error	0.04	0.56	0.04	0.55
Return	0.00	0.06	0.00	0.00
Offer	0.02	1.42	0.02	1.06
Auditor 20% Industry Share	0.06	7.26**		
Auditor Industry Share			0.08	10.70**
Average adjusted R^2	11.3%	4.86**	10.9%	5.12**

Regulated industries include all industries under the heading "Economic Regulation" in Table 1 of Weiss and Klass (1986) plus insurance and aerospace. For other definitions, see Table 1.

individual Big Six do not drive the positive association between our industry specialization measures and disclosure quality (not tabled). Our industry specialization variables remain significant in the unregulated sample and insignificant in the regulated sample with either overall or annual disclosure quality as the dependent variable.[19]

We also examine whether we are simply picking up an economy-wide effect. We measure each audit firm's economy-wide market share based on sales audited within Compustat. We then add this Compustat-based ranking of the Big Six to our regression model. This does not add significant explanatory power to our model nor does it change the significance of our industry-specialist variables (not tabled). It appears that our basic model indeed identifies an industry-specific phenomenon.

Finally, we split our data based on the median size of the firms in our sample, to test whether we are simply picking up some sort of non-linear size effect with our industry-specialization measure. Our measure of industry specialization is correlated with size and size is a significant determinant of disclosure quality (Lang and Lundholm, 1993). By splitting on client size we allow for potentially different size and disclosure quality relationships across our sample. We find that there is no material impact on the significance of our industry specialization variable (not tabled). It remains significant $(p < 0.01)$ in both size partitions although the *t*-statistics are slightly lower, possibly due to the smaller sample sizes. We also added the square of sales to the model to capture size and disclosure quality non-linearities, but it was insignificant and had no effect on our treatment variables.

Table 5. Awards for excellence/letters of commendation analysis.
The AIMR industry committees issue Awards for Excellence and Letters of Commendation to up to two firms in each industry that they perceive to exhibit outstanding disclosure practices. We estimate the following logistic regression model. We report the average coefficients from year-over-year logistic regressions and provide *t*-tests for whether they differ significantly from zero:

$$\text{Award} = \alpha_1 + \beta_1 \text{ Market value}_i + \beta_2 \text{ Earnings-return correlation}_i + \beta_3 \text{ Standard deviation returns}_i + \beta_4 \text{ Forecast error}_i + \beta_5 \text{ Return}_i + \beta_6 \text{ Offer}_i + \beta_7 \text{ Specialist}_i.$$

	Average Coefficient	*t*-statistic	Average Coefficient	*t*-statistic
Panel A: All Industries				
Intercept	−4.65	−6.32**	−4.70	−6.53**
Market value	1.58	5.65**	1.63	6.20**
Earnings-return correlation	0.60	1.68	0.63	1.77
Standard deviation returns	−0.01	−0.04	−0.02	−0.05
Forecast error	0.43	1.00	0.42	1.01
Return	0.05	0.09	0.03	0.07
Offer	0.48	3.17*	0.48	2.87*
Auditor 20% Industry Share	0.81	6.57**		
Auditor Industry Share			8.86	2.38†
Average adjusted Psuedo R^2	5.0%	4.56**	5.0%	5.18**
Panel B: Regulated Industries				
Intercept	−11.40	−2.50†	−11.19	−2.58*
Market value	4.39	2.08†	4.32	2.14†
Earnings-return correlation	−0.16	−0.22	0.32	0.61
Standard deviation returns	0.94	1.72	1.55	1.39
Forecast error	−0.70	−1.63	0.14	0.14
Return	1.69	1.39	1.97	1.38
Offer	2.61	1.46	2.76	1.55
Auditor 20% Industry Share	2.79	1.02		
Auditor Industry Share			1.64	0.98
Average adjusted Psuedo R^2	10.8%	3.92*	10.1%	4.49**
Panel C: Unregulated Industries				
Intercept	−4.49	−5.78**	−4.55	5.71**
Market value	1.15	5.23**	1.31	6.16**
Earnings-return correlation	0.76	2.29†	0.78	2.32†
Standard deviation returns	−0.45	−0.94	−0.40	−0.78
Forecast error	0.74	1.12	0.70	1.04
Return	−0.70	−0.15	−0.11	−0.25
Offer	0.50	1.86	0.48	1.57
Auditor 20% Industry Share	1.05	6.08**		
Auditor Industry Share			1.01	2.23†
Average adjusted Psuedo R^2	7.0%	6.78**	6.5%	5.53**

Award is equal to one if the client receives an Award for Excellence or Letter of Commendation in Corporate Reporting from the AIMR committees and zero otherwise. All other variables are defined in Table 1. Regulated industries include all industries included under the heading "Economic Regulation" in Table 1 of Weiss and Klass (1986) plus insurance and aerospace. For AIMR industries that include more than one two-digit SIC code, we categorize the AIMR industry based on the regulated/unregulated status of the majority of the firms within the AIMR industry. †, *, and ** indicate two-tailed significance at 0.10, 0.05, and 0.01.

4.2. Alternative Measures of Audit firm Industry Specialization

We consider a number of alternative specifications of audit firm industry specialization. The industry specialization literature typically uses measures of specialization based on sales, assets or number of firms audited. Industry specialization measures based on portion of sales or assets audited are highly correlated ($r = 0.95$) and yield almost identical results in our regression analyses using 20% of assets audited (not tabled). However, industry concentration based on the number of firms audited is not as highly correlated with sales or asset measures ($r = 0.56$) and does not generate significant results using a 20% cutoff. Hogan and Jeter (1999) note a similar difference in results when they use number of clients instead of assets audited to measure industry specialization in tests for an association between audit firm and client industry concentration levels.

We also examine different cut-off levels for industry specialization (results not tabled). Our basic finding of an association between industry specialization and disclosure quality (both overall and annual scores) holds at a 15% market share cut-off when based on sales or assets, but not at a 15% market share cut-off when based on the percentage of firms. The association is still positive but not significant at a 10% market share for sales, assets and firms. We also used two indicator variables to explore how a dual cut-off (greater than 20%, and fewer than 10%) performed. We observe a significant positive coefficient on the greater than 20% variable and a negative sign on the fewer than 10% variable for the unregulated industry sample in tests on the overall score.

We assess self-reported measures of industry specialization based on the information on the audit firms' web sites. Krishnan (2001) shared her data with us on self-reported specializations prior to the merger of Price Waterhouse and Coopers & Lybrand. The correlation between the self-reported specialization and our measure based on sales was 0.078. Not surprisingly, we find no association using these self-proclaimed measures. We believe the lack of association reflects the inherent problem with self-proclaimed specialization in that it may reflect a desire to gain market share and not necessarily reflect actual ability to deliver higher quality. However, when we use a measure based on both self-proclaimed specialization and a 20% market share, we document nearly identical results to when we use 20% market share alone.[20]

We also evaluate a measure designed to rule out the argument that just the most popular audit firms in each industry are designated as specialists, while audit firms with significant firm specific industry investments are systematically overlooked. For each two-digit SIC code, we calculate each audit firm's percentage of its total sales audited across all industries. We then rank audit firms within each industry based on this measure of the importance of each industry to a firm's overall practice.[21] The resulting measure produces very similar results to those we document in the paper. The fact that we only include Big Six audit firms in our sample explains the similarity. The size variation across the Big Six is relatively small, so the firm which audits the most sales in an industry tends to also derive the biggest proportion of its total audited sales from the industry relative to other firms.

5. Summary and Conclusions

We document an association between audit firm industry specialization and disclosure quality for firms in unregulated industries but not for firms in regulated industries. The results suggest that industry-specialist audit firms in unregulated industries provide value added services to their clients in the form of improved disclosure quality, and that the choice of an industry-specialist auditor is a signal of enhanced disclosure quality. Our results also suggest that audit firms have no impact on disclosure quality in regulated industries where regulators provide an additional layer of monitoring, there is less information asymmetry, and clients have less incentive and opportunity to provide enhanced disclosure quality.

In support of our improved disclosure quality findings, recent research suggests a link between industry specialization and audit quality in the form of higher quality earnings (Balsam et al., 2003; Gramling et al., 2000). While we believe audit quality plays a secondary role in our study, we expect the skills that enable an auditor to provide enhanced disclosure advice to clients to be highly correlated with improved audit quality. There remains opportunity in this area to extend our knowledge of the linkages between industry specialization and audit and earnings quality by examining our conjectures about differences in the industry specialist's role in regulated versus unregulated industries, and to examine the association between industry specialization and refined measures of earnings management and financial reporting quality.

Our research is subject to certain limitations. First, we rely on market share to capture industry specialization. While we believe increased market share should be associated with the underlying industry specialization construct, it is possible that firms with a small group of clients in the industry possess a high level of industry expertise and knowledge. Second, we implicitly treat industry specialization as an exogenous variable in our empirical analysis. Clearly, the decision to hire a specialist is an endogenous choice variable of a firm.[22] Our results are consistent with clients hiring specialist auditors as part of their overall disclosure strategy, and with clients not hiring specialists when they want to minimize information transfer to competitors. However, further research is needed to examine these conjectures. Research could address these issue by examining the characteristics of firms that select specialist auditors, and also by studying the characteristics of firms that switch to specialists from non-specialists or vice versa.

Our results suggest that the role of industry-specialists is different in regulated versus unregulated industries. Prior research has also documented differences in auditor concentration levels in regulated industries versus unregulated industries (Hogan and Jeter, 1999; Danos and Eichenseher, 1982). This suggests that future research should pay close attention to the impact of regulation on the auditor/client relationship and the hypothesized role of auditor industry-specialization.

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Notes

1. Enhanced disclosure quality is intended to represent disclosure beyond the minimum requirements of GAAP.
2. It is reasonable to hypothesize that auditors specialize on dimensions other than disclosure quality within regulated industries. For example auditors may specialize in aiding clients in complying with regulatory disclosure requirements. Such specialization may enable specialist auditors to develop significant cost advantages over non-specialists. In addition, Warfield et al. (1995) provide evidence that firm's in regulated industries engage in less earnings management than unregulated firms. Their theory also suggests a lesser role for disclosure quality in regulated industries.
3. We also implicitly align on source of capital (public equity markets) and to some degree size (large). It is quite reasonable to claim that Big Six firms specialize in providing services to large publicly traded clients and then further specialize by industry within the large public company market.
4. To test our conjecture, we used a sample of 3,501 firms who disclosed their audit and non-audit fees in their fiscal 2000 proxy statement. We defined industry specialization using the 20% of sales cut-off level in two-digit SIC industries based on fiscal 2000 Compustat data. The mean ratio of non-audit to total fees was 43.2% for non-specialist audited clients and 51.5% for specialist audited clients. The use of a ratio helps insure that size alone does not drive our comparison. Consistent with our expectations the mean for industry specialist auditors was significantly larger than non-specialist ($p < 0.0001$). Fee data was not publicly available for the same time period as our AIMR data. While we believe this evidence is consistent with our arguments, some caution should be exercised in interpreting this evidence given the time difference.
5. Enhanced disclosure is a form of voluntary disclosure. Accordingly, the costs to enhanced disclosure parallel the costs to voluntary disclosure. For a complete summary of voluntary disclosure trade-offs see Verechia (2001).
6. For evidence on industry specialists charging higher fees see Mayhew and Wilkins (2003), Ferguson and Stokes (2002), and Francis et al. (1995).
7. It is interesting to note that the SEC organizes its staff along industry monitoring lines. This suggests that the SEC considers industry specific reporting to be an important form of expertise.
8. The auditors' specialized knowledge on fulfilling specific reporting requirements for regulated industries (i.e., insurance companies' statutory filings) may provide them with important cost advantages over non-specialized auditors. This may explain why prior research has identified higher auditor industry concentration ratios in regulated industries (Hogan and Jeter, 1999; Danos and Eichenseher, 1982), but has found no evidence of audit fee premium in regulated industries (Palmrose, 1986; Pearson and Trompeter, 1994).
9. The 20% cut-off results in the following distribution of specialists: 30% of the industries have one specialist, 59% have two specialists and 11% have three specialists, where industries are defined by two-digit SIC codes over the timeframe of our study.

10. Appendix A of the 1991–1992 AIMR Corporate Information Committee Report provides a description of what the analysts consider in their evaluation of disclosure quality.
11. Overall disclosure quality scores are highly correlated with annual report scores and quarterly and other published information scores for industries that report scores for each category.
12. The rank of the highest value firm (e.g., largest firm) in the industry equals n and the smallest equals one.
13. Inferences from pooled time series data are not qualitatively nor quantitatively different.
14. Compustat does not report audit firm codes for companies in the financial industries; therefore, Auditor Industry Share for financial companies is the percentage of sales the client's audit firm audits in the client's industry for the companies in our sample.
15. We include two other industries (insurance and aerospace) not listed in Weiss and Klass (1986) in our regulated sample. Insurance companies are regulated at the state level and are required to file annual reports with state insurance commissioners. Aerospace companies engage in high levels of federal defense contracting. Most defense contracts are written on a cost plus basis that requires these firms to file financial reports with the department of defense. In both cases these industries face government monitoring that we expect to decrease the role of the auditor.
16. Hogan and Jeter's (1999) definition classifies the same industries as regulated that we do plus: coal mining, environment, food and beverage, and precious metals. They classify aerospace as unregulated, whereas we classify it as regulated. The results we report do not differ significantly from unreported results using Hogan and Jeter's definition.
17. Compustat does not include auditor codes for the full sample period for the insurance, banking, savings institutions, and financial services industries. We hand collected auditor codes for these industries from Compact Disclosure and based auditor industry specialization measures for these industries on the market share of our sample firms.
18. We specifically check for problems with multicollinearity in our analysis given the significant correlation between our treatment variables and control variables. None of our year-by-year regressions indicate any problem (VIF's are all less than two).
19. None of the individual audit firm indicator variables is significant nor do they increase the explanatory power of the model.
20. It appears auditors tend to grossly overstate the number of industries in which they specialize. The web-based disclosures list more industries than are supported by our measures of industry specialization.
21. This metric captures audit firms that derive a significant portion of their total revenue from an industry, even if the firm audits a smaller segment of that industry than a much larger auditor does. This contrasts with our reported measure that captures the auditor's market share in the industry without regard to the relative importance of the industry to individual firms' practices.
22. When we exclude the specialist variable from our regression the descriptive power of the model decreases by approximately 7% and there is little change in the control variables.

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